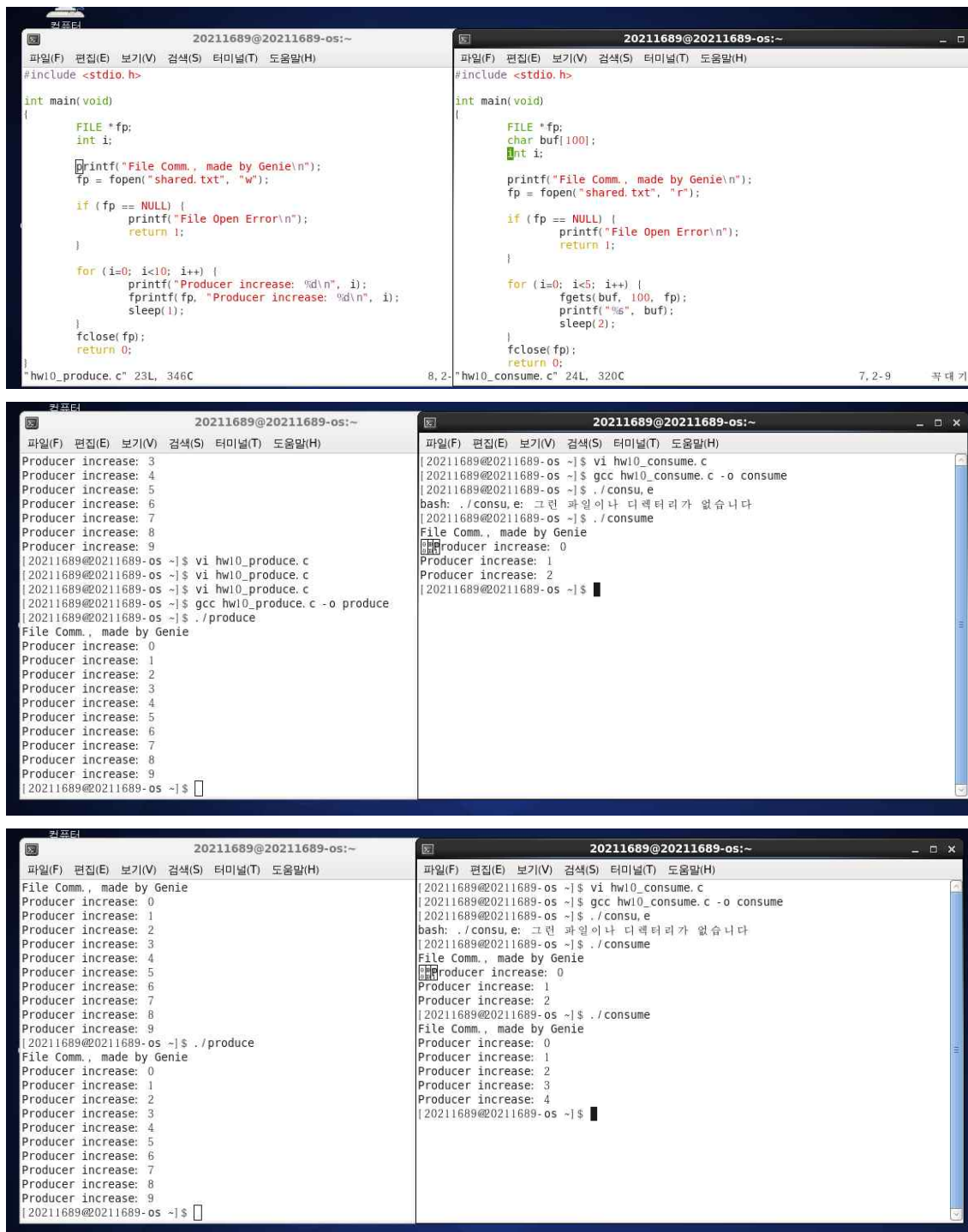
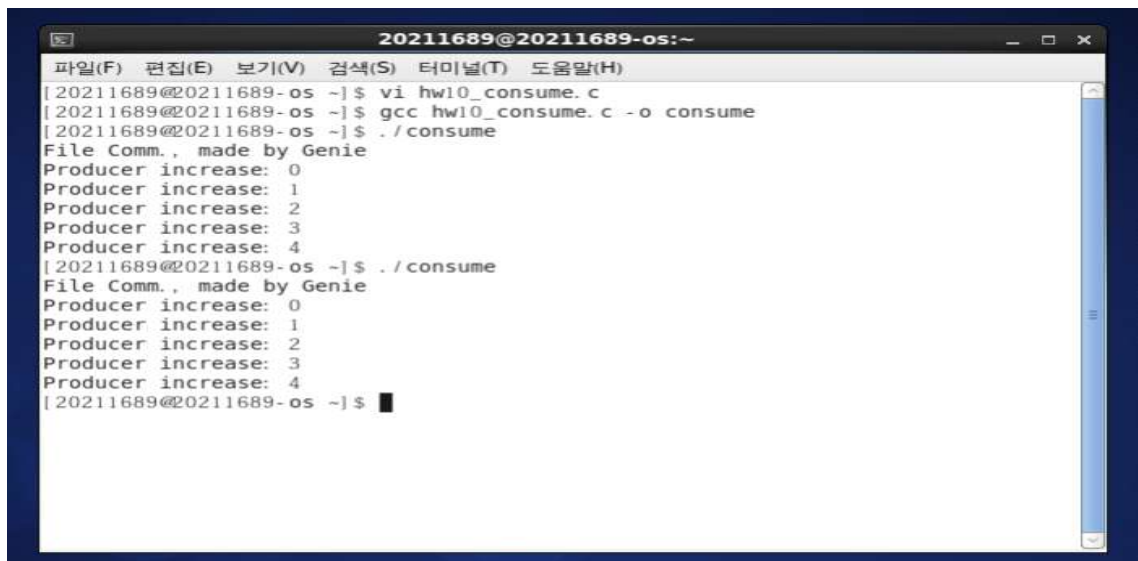


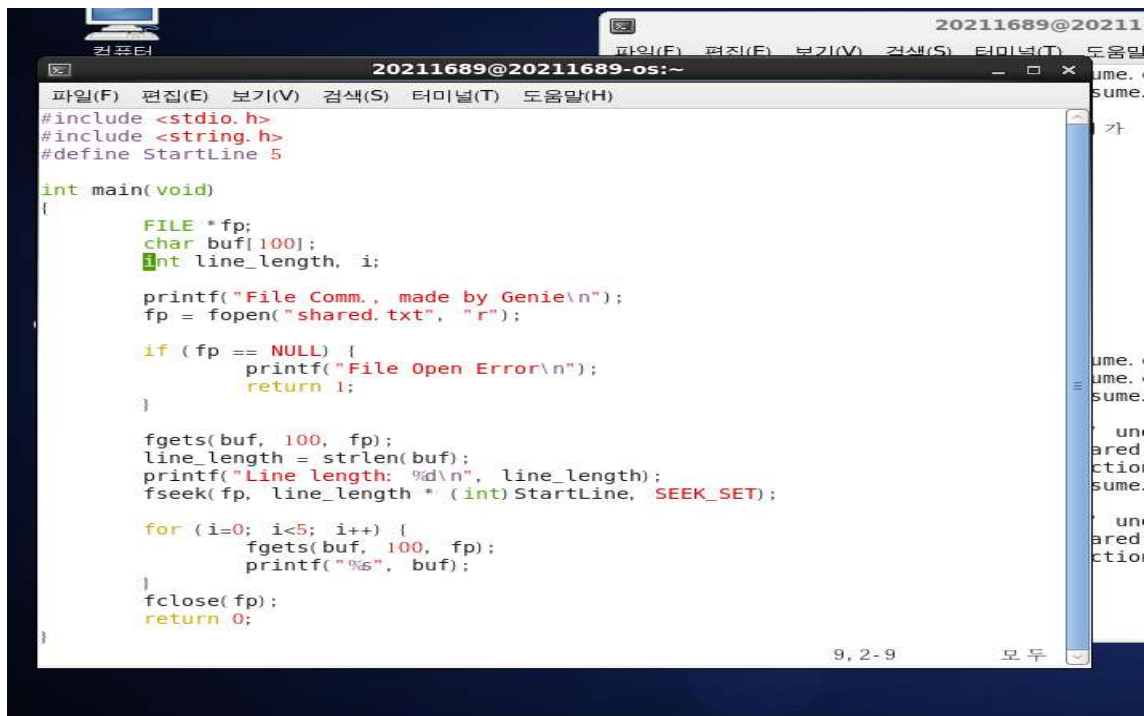


1. 생산자 프로그램과 소비자 프로그램을 동시에 실행하면 잘못된 결과가 출력되는 이유는 임계구역에 동시에 들어가서 동시 접속되었기 때문이다.



```
20211689@20211689-os:~
파일(F) 편집(E) 보기(V) 검색(S) 터미널(T) 도움말(H)
[20211689@20211689-os ~]$ vi hw10_consume.c
[20211689@20211689-os ~]$ gcc hw10_consume.c -o consume
[20211689@20211689-os ~]$ ./consume
File Comm., made by Genie
Producer increase: 0
Producer increase: 1
Producer increase: 2
Producer increase: 3
Producer increase: 4
[20211689@20211689-os ~]$ ./consume
File Comm., made by Genie
Producer increase: 0
Producer increase: 1
Producer increase: 2
Producer increase: 3
Producer increase: 4
[20211689@20211689-os ~]$
```

2. 파일을 read로 읽어오기 때문에 맨 앞에서부터 읽어서이다.



```
20211689@20211689-os:~
파일(F) 편집(E) 보기(V) 검색(S) 터미널(T) 도움말(H)
#include <stdio.h>
#include <string.h>
#define StartLine 5

int main(void)
{
    FILE *fp;
    char buf[100];
    int line_length, i;

    printf("File Comm., made by Genie\n");
    fp = fopen("shared.txt", "r");

    if (fp == NULL) {
        printf("File Open Error\n");
        return 1;
    }

    fgets(buf, 100, fp);
    line_length = strlen(buf);
    printf("Line length: %d\n", line_length);
    fseek(fp, line_length * (int)StartLine, SEEK_SET);

    for (i=0; i<5; i++) {
        fgets(buf, 100, fp);
        printf("%s", buf);
    }
    fclose(fp);
    return 0;
}
```

```
20211689@20211689-os:~
파일(F) 편집(E) 보기(V) 검색(S) 터미널(T) 도움말(H)
[20211689@20211689-os ~]$ ./produce
File Comm., made by Genie
Producer increase: 0
Producer increase: 1
Producer increase: 2
Producer increase: 3
Producer increase: 4
Producer increase: 5
Producer increase: 6
Producer increase: 7
Producer increase: 8
Producer increase: 9
[20211689@20211689-os ~]$ vi hw10_consu, e
[20211689@20211689-os ~]$ ls
a.out      codeAll    hw6_1.c   program1.c
backup     consume    hw6_2.c   run
backup2    gcc_test3  main.c    sample2
code1.c    hw10_consume.c  print.c   sample3
code2.c    hw10_produce.c print.o    shared.txt
code3.c    hw6.c      produce    test.txt
[20211689@20211689-os ~]$ vi hw10_consume.c
[20211689@20211689-os ~]$ vi hw10_consume.c
[20211689@20211689-os ~]$ ./consume
File Comm., made by Genie
Line length: 21
Producer increase: 5
Producer increase: 6
Producer increase: 7
Producer increase: 8
Producer increase: 9
[20211689@20211689-os ~]$
```

3. #define StartLine 5 코드를 추가하고, fgets로 줄을 읽고, 길이를 알아내서 fseek()가 길이를 계산해서 이동한다. 실행시켰을 때 5부터 출력이 된다.